

DUAL PORT RAM–REPORT USING VERILOG HDL ON ARTIX-7 FPGA

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INTRODUCTION

In modern digital systems, memory access speed plays a crucial role in overall system performance. Traditional single-port RAM allows only one read or write operation at a time, which can cause delays in multi-processor or parallel processing systems.

Dual-Port RAM (DPRAM) overcomes this limitation by providing two independent ports that can access the same memory simultaneously. Each port can perform read or write operations independently, enabling parallel data access and reducing memory bottlenecks.

In this project, a simple dual-port RAM is designed using Verilog HDL, where: - One port is used for writing data into memory - Two independent ports are used for reading data from memory - All operations are synchronized with the system clock

KEY FEATURES OF DUAL-PORT RAM

- Simultaneous access to memory through two independent ports
 - Improved performance due to parallel read operations
 - Reduced waiting time when accessing different memory locations
 - Suitable for multi-processor and embedded systems
 - Synchronous operation using a clock signal
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DESIGN SPECIFICATION

- Memory Size: 4 locations
- Data Width: 1 bit
- Address Width: 2 bits
- Write Operation: Synchronous (clock-based)
- Read Operation: Synchronous
- Technology Used: Verilog HDL

BLOCK DIAGRAM DESCRIPTION

The dual-port RAM consists of: - A memory array of 4 locations - One write enable signal (we)
- One write address input (waddr) - Two read address inputs (raddr_a and raddr_b) - Two data outputs (a and b) - A common clock signal

When the write enable is active, data is written into the memory at the specified write address. At the same time, data from two different memory locations can be read using the two read ports.

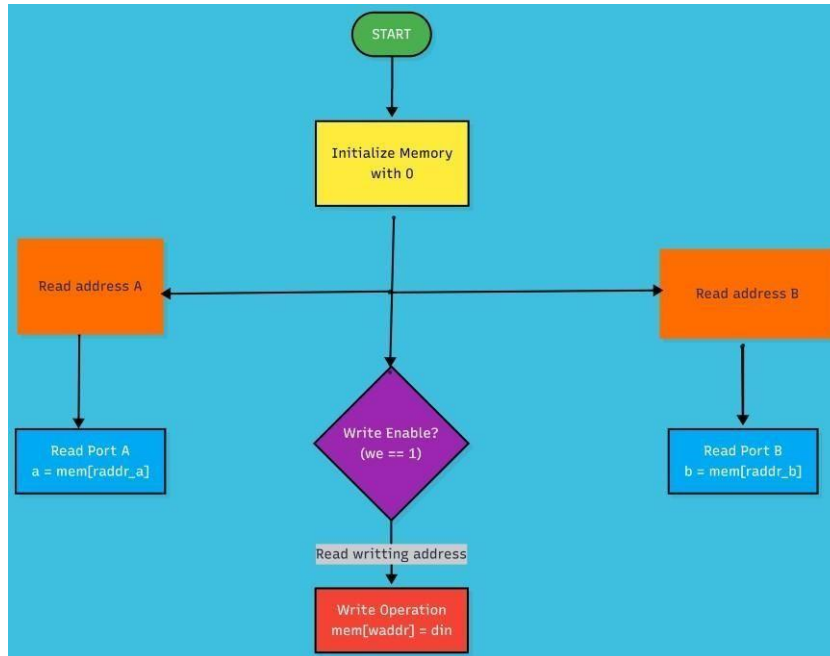
ALGORITHM

1. Initialize all memory locations to zero.
 2. Wait for the rising edge of the clock.
 3. If write enable (we) is high:
 - Write input data (din) into memory at address waddr.
 4. Read data from memory location raddr_a and assign it to output a.
 5. Read data from memory location raddr_b and assign it to output b.
 6. Repeat the process for every clock cycle.
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BLOCK DIAGRAM



FLOW CHART



VERILOG CODE

```

`timescale 1ns / 1ps
module dual ( input
clk, input we, input
din, input [1:0] waddr,
input [1:0] raddr_a,
input [1:0] raddr_b,
output reg a, output
reg b ); reg mem [3:0];
integer i ; initial begin
for (i = 0; i < 8; i = i +
1) mem[i] = 8'b0; end
always @(posedge clk)
begin
if (we)
mem[waddr] <= din;
a <= mem[raddr_a];

```

```
b <= mem[raddr_b];  
end endmodule
```

VERILOG CODE ANALYSIS Memory

Declaration

A register array is used to represent memory with four locations.

Initialization Block

All memory locations are initialized to zero using a for loop. This ensures predictable output during simulation. **Always Block**

The always block is triggered on the positive edge of the clock: - If write enable is high, data is written into memory - Read operations occur simultaneously for both read ports This ensures synchronous operation of both read and write processes.

TESTBENCH

```
`timescale 1ns / 1ps  
module dualtb;  
  reg clk; reg we;  
  reg din; reg [1:0]  
  waddr; reg [1:0]  
  raddr_a; reg [1:0]  
  raddr_b; wire a;  
  wire b;  
  dual uut (clk, we, din, waddr, raddr_a, raddr_b, a,  
  b); always #5 clk = ~clk ; initial begin  
  clk = 0; we = 0; din = 0; waddr = 0; raddr_a = 0; raddr_b = 0; #10; we  
  = 1;  
  waddr = 2'b00; din = 1'b0; #10;  
  waddr = 2'b01; din = 1'b1;  
  #10; waddr = 2'b10; din =  
  1'b0; #10; waddr = 2'b11; din  
  = 1'b1; #10; we= 0; raddr_a =  
  2'b00; raddr_b = 2'b01;  
  #10;
```

```
Raddr_a = 2'b10;  
Raddr_b = 2'b01;  
#10;  
#20; End  
endmodul  
e
```

TESTBENCH DESCRIPTION

The testbench is used to verify the functionality of the dual-port RAM design. [Testbench](#)

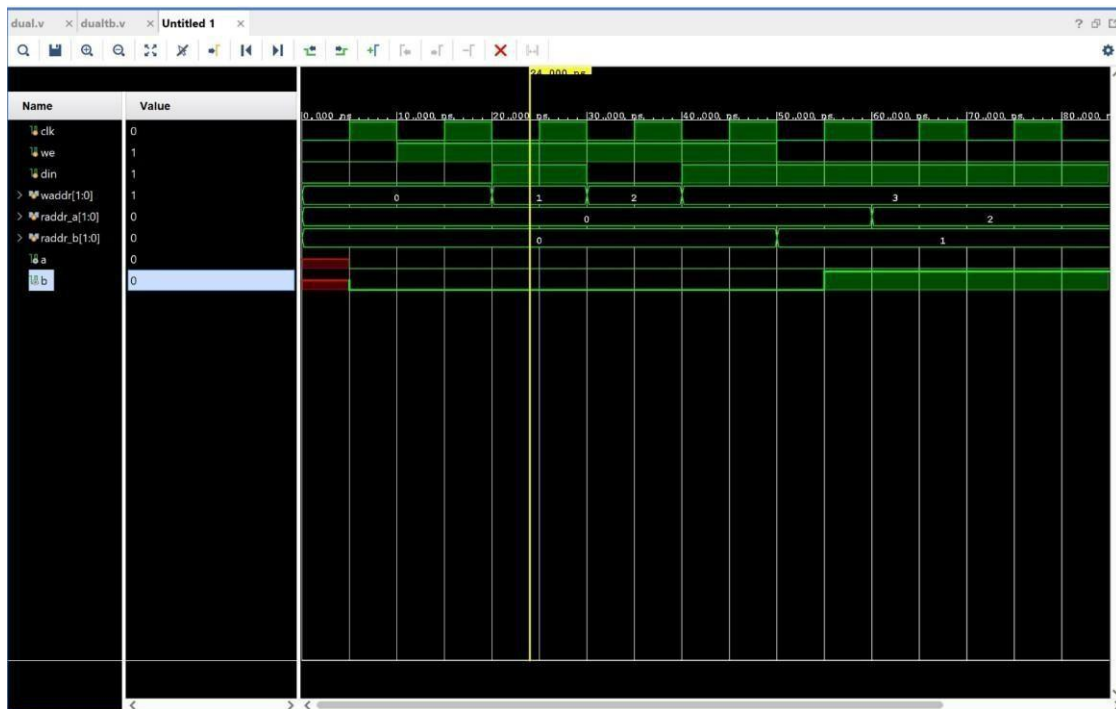
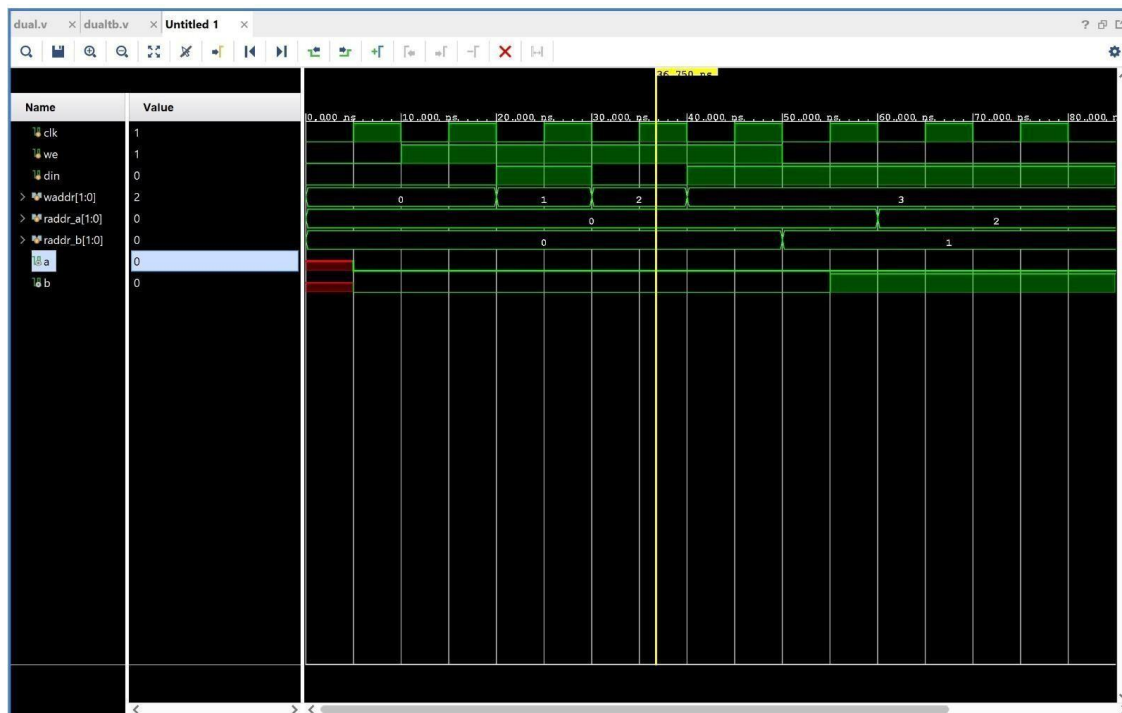
Operations

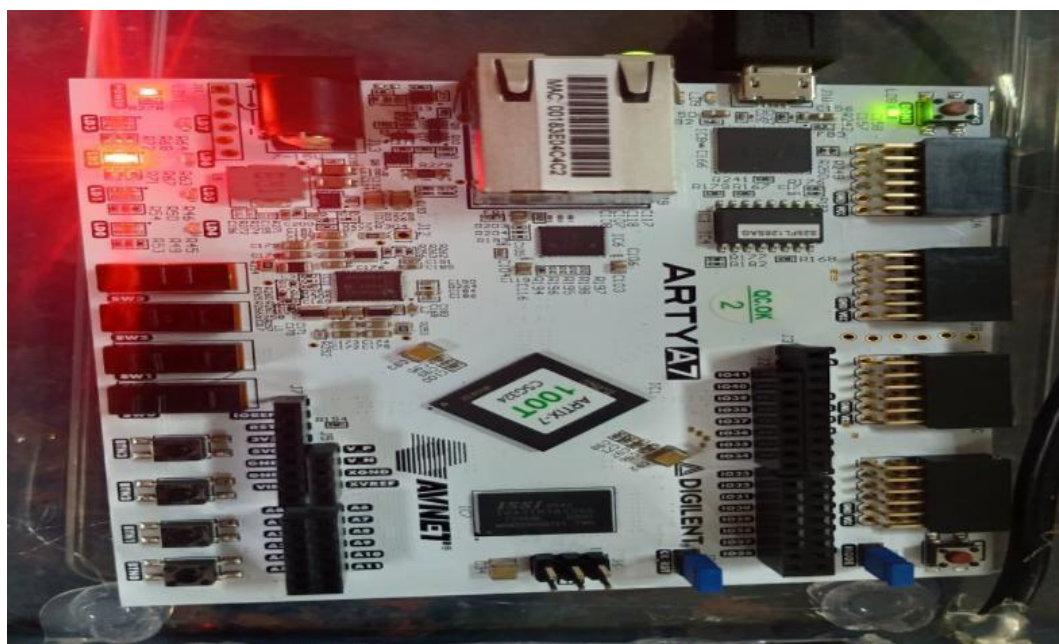
- Generates a clock signal with a 10 ns period
- Writes data into all memory locations sequentially
- Disables write mode
- Reads data from different memory locations simultaneously using two read ports

The testbench confirms that the RAM correctly supports parallel read operations while maintaining data integrity.

SIMULATION RESULTS

The simulation results show that: - Data is correctly written into memory when write enable is active - Both read ports can access different memory locations simultaneously - No data conflict occurs when accessing different addresses This verifies the correct operation of the dual-port RAM.





APPLICATIONS

- Multi-processor systems
- FIFO buffers
- Image and signal processing systems
- Communication systems
- Embedded systems

ADVANTAGES

- Faster memory access
- Supports parallel processing
- Efficient use of hardware resources
- Reduced system latency

CONCLUSION

In this project, a dual-port RAM was successfully designed and implemented using Verilog HDL. The design allows simultaneous access to memory through two independent read ports, improving system performance. The simulation results validate the correct functionality of the design, making it suitable for use in high-performance digital systems.

SYNTHESIS REPORT

```

-----
Vivado v2025.1 (64-bit)
SW Build 6140274 on Thu May 22 00:12:29 MDT 2025
IP Build 6138677 on Thu May 22 03:10:11 MDT 2025
SharedData Build 6139179 on Tue May 20 17:58:58 MDT 2025
Start of session at : Thu Jan 1 20:54:07 2026
Process ID : 13492
Current directory : C:/Users/ambig/Vivado/project_5.runs/synth_1
Command line :
vivado.exe -log dual.vds -product Vivado -mode batch -messageDb vivado.pb
-notrace -source dual.tcl
Log file : C:/Users/ambig/Vivado/project_5.runs/synth_1/dual.vds
Journal file : C:/Users/ambig/Vivado/project_5.runs/synth_1/vivado.jou
Running On : LAPTOP-3SUQPA02
Platform : Windows Server 2016 or Windows 10
Operating OS : 26200
Processor : 13th Gen Intel(R) Core(TM) i5-13500H
CPU Frequency: 3187 MHz
Physical Cores: 12
Logical Cores : 16
Host Memory : 16852 MB
Swap Memory : 1879 MB
Total Virtual: 18731 MB
Available Virtual: 3691 MB
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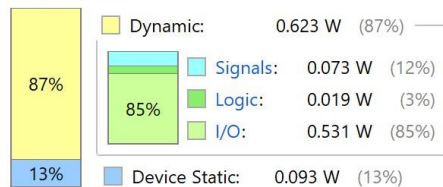
```

Summary

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power: 0.717 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 28.3°C
 Thermal Margin: 56.7°C (12.3 W)
 Ambient Temperature: 25.0 °C
 Effective θ_{JA} : 4.6°C/W
 Power supplied to off-chip devices: 0 W
 Confidence level: Low
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	inf	Worst Hold Slack (WHS):	inf	Worst Pulse Width Slack (WPWS):	NA
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	NA
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	NA
Total Number of Endpoints:	20	Total Number of Endpoints:	20	Total Number of Endpoints:	NA

There are no user specified timing constraints.